

Math 162

Problem set 1 solutions

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1. By hypothesis, for any $\epsilon > 0$ there exist $\delta_1, \delta_2 > 0$ such that for any tagged partition (P, τ) of $[a, b]$, the inequality $\|P\| < \delta_1$ implies that $|S(f, P, \tau) - I| < \frac{\epsilon}{2}$, and $\|P\| < \delta_2$ implies that $|S(f, P, \tau) - I| < \frac{\epsilon}{2}$. Let $\delta := \min(\delta_1, \delta_2)$, and choose any tagged partition (P, τ) such that $\|P\| < \delta$ (for example, one could take P to be the equipartition into n subintervals where $\frac{1}{n} < \frac{\delta}{b-a}$, and τ to be given by the right-hand rule). Then $\|P\| < \delta \leq \delta_1, \delta_2$, which implies that

$$\begin{aligned} |I - I'| &= |I - S(f, P, \tau) + S(f, P, \tau) - I'| \\ &\leq |S(f, P, \tau) - I| + |S(f, P, \tau) - I'| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Since ϵ was arbitrary, it follows that $I = I'$.

2. (a) Suppose that f is Riemann integrable with $I := \int_a^b f(x) dx$, and fix $\epsilon > 0$. Then there exists $\delta > 0$ such that for any tagged partition (P, τ) satisfying $\|P\| < \delta$, we have $|S(f, P, \tau) - I| < \epsilon$. Let $N := \frac{b-a}{\delta}$. Taking P to be the equipartition into n subintervals, we have $\|P\| = \frac{1}{n}(b-a)$, and hence $n > N$ implies that $\|P\| < \frac{1}{N}(b-a) = \delta$. Now if τ is the tagging given by the right-hand rule, we have $S_n(f) = S(f, P, \tau)$, and hence

$$|S_n(f) - I| = |S(f, P, \tau) - I| < \epsilon$$

as desired.

- (b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be the indicator function of the rationals:

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

We saw in class that f is not Riemann integrable, but we claim that f is fake Riemann integrable. Namely, since $x_i = \frac{i}{n} \in \mathbb{Q}$, we have $f(x_i) = 1$ for all $n \in \mathbb{N}$ and $1 \leq i \leq n$. It follows that

$$S_n(f) = \sum_{i=1}^n f(x_i) \Delta x = n \Delta x = 1$$

for all $n \in \mathbb{N}$, so for any $\epsilon > 0$ we have $|S_n(f) - 1| = 0 < \epsilon$ for all $n \in \mathbb{N}$ (thus any $N > 0$ works, e.g. $N := 1$).

3. Let $I := \int_a^b f(x) dx$ and fix $\epsilon > 0$. We must show that there exists $\delta > 0$ such that for any tagged partition (P, τ) satisfying $\|P\| < \delta$, we have $|S(g, P, \tau) - I| < \epsilon$. By assumption, there exists $\delta_1 > 0$ such that $\|P\| < \delta_1$ implies that $|S(f, P, \tau) - I| < \frac{\epsilon}{2}$.

Let $\delta_2 := \frac{\epsilon}{4|g(c) - f(c)|}$ (if $f(c) = g(c)$ then f and g are the same function and there is nothing to prove, so we can assume that $f(c) \neq g(c)$). Given a tagged partition (P, τ) such that $\|P\| < \delta_2$, we claim that $|S(g, P, \tau) - S(f, P, \tau)| < \frac{\epsilon}{2}$. If $c \notin \tau$ then we have $S(f, P, \tau) = S(g, P, \tau)$ and the difference is 0. If $c \in \tau$, then there exists $1 \leq j \leq n$ such that either

$$S(g, P, \tau) - S(f, P, \tau) = (g(c) - f(c)) \Delta x_j$$

or

$$S(g, P, \tau) - S(f, P, \tau) = (g(c) - f(c))(\Delta x_{j-1} + \Delta x_j)$$

(the latter case accounts for the possibility that $x_{j-1}^* = x_j^* = c$; in this case $j > 1$). In either case we have

$$|S(g, P, \tau) - S(f, P, \tau)| \leq |g(c) - f(c)| \cdot 2\|P\| < 2|g(c) - f(c)|\delta_2 = \frac{\epsilon}{2}$$

as claimed.

Finally, put $\delta := \min(\delta_1, \delta_2)$. If (P, τ) is such that $\|P\| < \delta$, then $\|P\| < \delta_1, \delta_2$ and hence

$$\begin{aligned} |S(g, P, \tau) - I| &= |S(g, P, \tau) - S(f, P, \tau) + S(f, P, \tau) - I| \\ &\leq |S(g, P, \tau) - S(f, P, \tau)| + |S(f, P, \tau) - I| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$